

Advanced Econometrics – Lab 06 – Mock test**Exercise 1**

Adrian Barnett, Inger Mewburn, and Sara Schorter in their article¹ "Working 9 to 5, not the way to make an academic living: observational analysis of manuscript and peer-review submissions over time" presented a proof how difficult is to be an academician [yes, it is, you can believe us! ;-)].

"The «publish or perish» mantra is well established in academic circles, but do researchers have to sacrifice leisure time to achieve this ideal? [...] The pressure to publish creates a vicious cycle whereby publishing more papers has the potential to push the university up in the rankings, which, in turn, puts more pressure on academics to keep working to maintain their university's place in the rankings (and perhaps their own jobs). At the same time, universities in many countries have seen a massive increase in student numbers and teaching loads that squeeze the available time for research.

Anecdotally at least, academics and medical students are working longer hours to accommodate the demands of research and publishing.

The objective of the study was to determine whether researchers are submitting manuscripts and peer-reviews to BMJ journals out of hours and whether this has changed over time. The data consisted of info on research manuscripts submitted to the British Medical Journal (BMJ) between 2012 and 2019 for which an author's address could be geocoded. The aim of the study is to confirm the hypothesis that considerable amount of submissions takes place on weekends, on national holidays, and out of hours (early mornings and late nights)."²

Your lecturers decided to repeat this analysis with econometric models. What is more, your lecturers decided to study the number of submissions by hours. For this reason, we aggregated the data over hours for each country. As a result, we received counts, that is numbers of submissions per each hour per each country. Page 2 presents output from a Poisson model for those counts.

- a) Interpret the estimate of the parameter for *hour12* in the Poisson model presented below.
- b) What hour do researchers submit the most papers? Interpret the estimate's value.
- c) What hour is connected with the least number of submitted papers? Interpret the value.
- d) Are all the variables jointly significant?
- e) What other econometric models could be estimated for the data? Explain.

¹Barnett Adrian, Mewburn Inger, Schroter Sara. Working 9 to 5, not the way to make an academic living: observational analysis of manuscript and peer-review submissions over time BMJ 2019; 367 :l6460

²Barnett Adrian, Mewburn Inger, Schroter Sara, p. 1.

Call:

```
glm(formula = n ~ as.factor(hour), family = poisson, data = TableOfCounts)
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Deviance Residuals:

Min	1Q	Median	3Q	Max
-14.663	-7.160	-2.870	0.008	57.321

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	3.72759	0.02516	148.169	< 2e-16 ***
as.factor(hour)1	-0.52250	0.04123	-12.672	< 2e-16 ***
as.factor(hour)2	-1.05345	0.04948	-21.292	< 2e-16 ***
as.factor(hour)3	-1.58185	0.06638	-23.830	< 2e-16 ***
as.factor(hour)4	-1.84838	0.07456	-24.791	< 2e-16 ***
as.factor(hour)5	-2.03427	0.07988	-25.468	< 2e-16 ***
as.factor(hour)6	-1.79517	0.07182	-24.996	< 2e-16 ***
as.factor(hour)7	-1.51261	0.06445	-23.469	< 2e-16 ***
as.factor(hour)8	-1.02511	0.04995	-20.522	< 2e-16 ***
as.factor(hour)9	-0.23029	0.03781	-6.090	1.13e-09 ***
as.factor(hour)10	0.38500	0.03261	11.805	< 2e-16 ***
as.factor(hour)11	0.77368	0.03041	25.440	< 2e-16 ***
as.factor(hour)12	0.89944	0.02984	30.143	< 2e-16 ***
as.factor(hour)13	0.82545	0.03017	27.361	< 2e-16 ***
as.factor(hour)14	0.87573	0.02994	29.247	< 2e-16 ***
as.factor(hour)15	1.02804	0.02931	35.070	< 2e-16 ***
as.factor(hour)16	1.04376	0.02925	35.680	< 2e-16 ***
as.factor(hour)17	1.00189	0.02942	34.059	< 2e-16 ***
as.factor(hour)18	0.73011	0.03062	23.841	< 2e-16 ***
as.factor(hour)19	0.39614	0.03254	12.174	< 2e-16 ***
as.factor(hour)20	0.11412	0.03461	3.298	0.000975 ***
as.factor(hour)21	0.17065	0.03415	4.996	5.84e-07 ***
as.factor(hour)22	0.23522	0.03366	6.988	2.79e-12 ***
as.factor(hour)23	0.20938	0.03386	6.184	6.24e-10 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

Null deviance: 105214 on 876 degrees of freedom
 Residual deviance: 79954 on 853 degrees of freedom
 AIC: 84146

Number of Fisher Scoring iterations: 6